Computer Vision

Logesh. S

35/

255 − 50 = 205

255 − 100 = 155

255 − 150 = 105

255 − 200 = 55.

1.)

(b) Negative transformation

$S = 255 - r$

Orginal pixel	Transformed Pixel
50	205
100	155
150	105
200	55

(a) Grey level transformation in image Enhancement

IT is used to improve the image appearance by changing pixel intensity values.

(2) Histogram Equilization.

Intensity	frequency.
0	4
1	6
2	10
3	6

PDF ?

formula ?

(b) CDF = ?

Intensity	frequency	CDF
0	4	4
1	6	10
2	10	20
3	6	26

(c) Equalized intensity Value

$$S = \frac{CDF}{26} \times 3$$

Intensity	Calculation
0	$\frac{4}{26} \times 3 = 0.46$
1	$\frac{10}{26} \times 3 = 1.5$
2	$\frac{20}{26} \times 3 = 2.30$
3	$\frac{26}{26} \times 3 = 3$

(d) Histogram Equilization

IT improves image constrast by spreading pixel intensity value over the full intensity range.

3.)

(a) Gaussian filtering is prefered over averaging filtering.

IT preserves better edge when compared to average filter.

IT Gives The weight to The centre

IT Reduces The noises.

(b) Apply Gaussian Kernel

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$= \frac{1}{16}$

Image region

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

$$= \frac{1}{16}(10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10)$$

$$= \frac{1}{16}(360)$$

$$= \frac{360}{16} = 22.5$$

$$= 22.5 \approx 23.$$

$$\begin{array}{r} 22.5 \\ 16\overline{)360} \\ 32 \\ \hline 40 \\ 32 \\ \hline 80 \\ 80 \\ \hline 0 \end{array}$$

16
16
32
3 | 1635
80

(4)

(a) Concept of edge detection.

Edge dection is nothing but it identifies object boundaries by dectecting sudden intensity changes.

(b) Horizontal Sobel mask.

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

image.

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$$\text{Gradient} = (-1 \times 10) + (-2 \times 10) + (-1 \times 10) + (1 \times 30) + (2 \times 30) + (1 \times 30)$$

$$= -10 - 20 - 10 + 30 + 60 + 30 = -40 + 120 = 80$$

Gradient = 86.

5) (b) Linear Contrast stretching to map the range [50,150] to [0,255]

r: 50, 100, 150.

$$\boxed{s = \cancel{\frac{r-50}{150}}} \quad s = \frac{r-50}{100} \times 255$$

When $r = 50$

$$s = \frac{50-50}{100} \times 255$$

$s = 0$.

When $r = 100$

$$s = \frac{100-50}{100} \times 255.$$

$$s = \frac{\cancel{50}}{2\cancel{100}} \times 255$$

$$s = \frac{225}{2}$$

$s = 112.5$

$s = 113$.

Original Value	New value
50	0
100	113
150	255.

$$2\overline{)225} = 112.5$$

When $r = 150$

$$s = \frac{150-50}{100} \times 225$$

$$= \frac{100}{100} \times 225$$

$s = 255$

(a) Constrast stretching

This will increases the DIB dark and bright pixels and making the image more details and visible